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THE GREAT COMPRESSION: THE WAGE STRUCTURE IN THE UNITED STATES AT MID-CENTURY*

CLAUDIA GOLDIN AND ROBERT A. MARGO

The era of wage stretching has been a current focus, but we direct attention here to a decade of extraordinary wage compression—the 1940s. Wages narrowed by education, job experience, and occupation, and compression occurred within cells. The 90–10 differential in the log of wages for men was 1.45 in 1940 but 1.06 in 1950. By the late 1980s it returned to its 1940 level, thus restoring a dispersion of 50 years ago. World War II and the National War Labor Board share some credit for the Great Compression, but much was due to an increased demand for unskilled labor when educated labor was greatly expanding.

The Great Compression—our term for the narrowing of the wage structure in the 1940s—followed the Great Depression of the 1930s and produced a wage structure more equal than that experienced since. We explore here the features of the narrowing wage structure of the 1940s, its quantitative significance compared with the recent widening of the post-1960s, and its determinants. We seek to resolve whether the 1940s were a unique decade of narrowing or whether the decade brought to the fore a set of structural changes that had been maturing for the previous half century. In the process, we explore hourly earnings of white-collar workers to determine whether the wage structure in 1940 was

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anomalous, and we look at industrial wage distributions to determine the impact of World War II.

The wage structure narrowed in many ways during the 1940s. The variance in the log of weekly wages, for male wage and salary earners working more than 39 weeks in 1939, was 0.325 in 1940, but 0.259 in 1950 (see Table I, column 2). The difference in the log of weekly wages between those at the ninetieth and those at the tenth percentiles, among all males, was 1.45 in 1940 but 1.18 in 1950. Educational and skill premiums plummeted, geographical differences in wages were reduced, and the variance of wages within these groups fell substantially. When the United States emerged from war and depression, it had not only a considerably lower rate of unemployment, it also had a wage structure more egalitarian than at any time since. Further, the new wage structure remained somewhat intact for several decades. According to Lester Thurow, writing in 1975: "After the wage differentials of the Great Depression and World War II had become embedded in the labor market for a number of years, they became the new standard of

TABLE I
SUMMARY MEASURES OF THE WAGE STRUCTURE, 1940 TO 1985

	(1) White men Log of weekly wages			(2) All men Log of weekly wages			(3) Men > 21 years, 35+ hours Log of weekly wages
	90-10	90-50	50-10	90-10	90-50	50-10	90-10
1940	1.414	0.674	0.740	1.447	0.654	0.793	1.449
1950	1.060	0.472	0.588	1.181	0.504	0.677	1.060
1960	1.211	0.521	0.690	1.250	0.567	0.723	1.157
1960a							1.15
1970							1.18
1980							1.32
1985							1.46
	Variance of the log of weekly wages						
1940		0.313			0.325		
1950		0.241			0.259		
1960		0.262			0.275		

Notes. Unless otherwise noted, the sample includes white males, 18 to 64 years old who were wage and salary earners working more than 39 weeks and earning more than one half the minimum wage on a full-time basis. 90-10 is the log weekly wage at the ninetieth percentile minus the log weekly wage at the tenth percentile. Similar definitions hold for the group above the median (90-50) and below (50-10). Column (3) is defined for comparability with Juhn, Murphy, and Pierce [1989]. Figure 1 graphs column (3). In this table, as in the others, the years 1940, 1950, and 1960 are those of the census, while the earnings data refer to 1939, 1949, and 1959.

Sources. 1940, 1950, and 1960 Public Use Microdata Samples (PUMS). Precise details concerning the sample can be obtained from the authors. The 1960a, 1970, 1980, and 1985 data are from Juhn, Murphy, and Pierce [1989, table 1].

relative deprivation and were regarded as 'just' even after the egalitarian pressures of World War II had disappeared. Basically, the same differentials exist to this day, thirty years later" [1975, p. 111].

The same differentials do not exist to *this* day. Rather, the movement toward equality in the 1940s was reversed in the post-1970 period. In fact, remarkable similarities exist between the narrowing wage structure of the 1940s and the widening wage structure after 1970. Comparing the two periods, as we do in Table I (column 3), conveys a sense of the level of inequality and the magnitude of decline in the 1940s. The difference between the log of weekly wages at the ninetieth and the tenth percentiles in 1940 was virtually identical to that in 1985 of 1.46. For 1950 the same statistic is 1.06, and probably no year since has produced greater equality of wages. In terms of this particular measure of the wage structure, inequality in 1940 was similar to that in 1985, and the change from 1940 to 1950 was substantial by recent standards. Inequality of income, however, should also include the unemployed. In terms of that measure, 1940 would be more unequal than 1985, although the decrease in inequality to 1950 would be even greater.¹

The wage structure, as can be seen in Figure I, has been on a long-run roller coaster ride since 1940—with inequality falling precipitously during the 1940s, rising slightly in the 1950s and 1960s, and finally increasing sharply from the 1970s.² The statistical properties of the initial fall and recent rise are, in many ways, mirror images of each other. Not only did the between-group variance change in comparable but opposite ways, but the decrease in the within-group variance in the 1940s was of similar magnitude to the increase in the post-1960s period.³

But here the similarity between the two periods ends. Scholars have been grappling with a wide range of supply and demand factors to explain the recent widening of the between- and within-group variances (Blackburn, Bloom, and Freeman [1990], Bound

1. Unemployment in 1939, the year to which the 1940 census refers, was 17.2 percent of the civilian labor force, and in 1940 it had declined to 14.6 percent [U. S. Bureau of the Census, 1975, series D-86]. Note that the unemployment figures include individuals on work relief. In 1949 unemployment was 5.9 percent of the civilian labor force.

2. The roller coaster analogy may be more appropriate to describe the time series of the returns to education, since the returns decreased in the 1940s, rose somewhat in the 1950s, fell in the 1970s, and then rose sharply in the 1980s.

3. The results on the explained and residual variance (see Table V) indicate that a somewhat larger fraction can be explained in the past. When we exclude certain variables (e.g., regions, marital status), the results are similar to those for the more recent period.

Log(Wage), 90th-10th

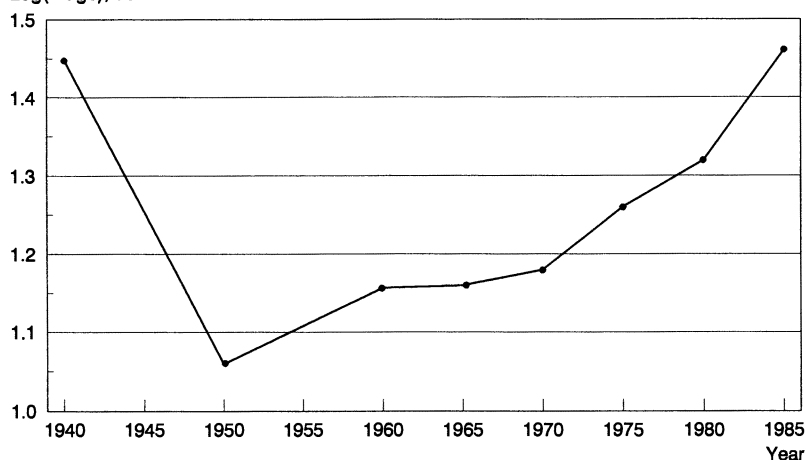


FIGURE I

Wage Dispersion across the Past Half Century: Difference in the Log Wage at the Ninetieth and Tenth Percentiles, 1940 to 1985

Source. Table I.

and Johnson [1989], Katz and Murphy [1992], Katz and Revenega [1989]; union and minimum wage factors are also considered in much of this literature). For the 1940s, however, unique historical events, institutional changes, and the ending of the Great Depression might explain much of the narrowing between and within groups. The 1930s and the 1940s were times of remarkable change in the economy, in the structure of wages in particular, and much may have been engineered by the government through the National Industrial Recovery Act (NIRA) and, during the war, the National War Labor Board (NWLB). Further, unemployment during the 1930s might have affected the wage structure. We explore both the residual influence of the depression and the impact of the war, in addition to the more usual influences of changing supply of and demand for skills, to resolve why the wage structure narrowed so quickly and why it remained compressed until the 1960s.

We are not the first to identify a compression in the wage structure in the 1940s.⁴ But previous work was hampered because

4. See also Kuznets [1953] and Goldsmith [1967] on earnings. The earnings structure, as measured by the proportion earned by the richest 5 percent, began to decline during the 1930s and narrowed far more substantially during the 1940s.

published census data on earnings are not comparable across census dates [Bartlett, 1978; Miller, 1955, 1958, 1966]. We make use here of the recently available public use microdata samples (PUMS) of the 1940 and 1950 censuses, along with the public use sample of the 1960 Census. These data permit a study of changes in the wage structure, at a highly disaggregated level, and enable us to make consistent links with more recent measures.

The reasons we offer for the substantial narrowing in the wage structure during the 1940s combine the short-run impact of wartime demand for less-skilled labor and the manner in which wartime wage changes were administered, with longer-run changes in the demand for and supply of educated workers in the 1940s. If the decade of the 1980s reinforced the rust belt, then surely the 1940s created the steel belt. The nation's relative demand for less-educated workers soared in the 1940s in comparison with later decades. At the same time the relative supply of highly educated workers rose. The initial impact was to compress the wage structure. In the longer run the continued increase in the supply of educated workers served to hold the new equilibrium in place. Various factors changed in the 1970s, and the wage structure eventually expanded. That, however, is a more recent chapter in the history of the wage structure (see Katz and Murphy [1992]).

I. THE WAGE STRUCTURE, 1940 TO 1960: EVIDENCE FROM THE CENSUS PUBLIC USE SAMPLES

A. *Between and Within Variance in the Wage Structure*

Our sample from the 1940, 1950, and 1960 PUMS consists of men between the ages of 18 and 64 employed for at least 40 weeks in the year preceding the census, who were wage or salary earners, and who earned, on average, more than one half the minimum wage on a full-time basis.⁵ In most cases our sample consists of

5. We have used the 1/1000 sample for 1940 and 1960, but the 1/100 sample for 1950 for which only sample-line individuals have wage data. We imputed earnings for those who were top-coded at 1.4 times the top code, although there were very few individuals who were top-coded in the 1940, 1950, and 1960 PUMS. One half the minimum wage in each of the three years is \$6, weekly, for 1940, \$8 for 1950, and \$20 for 1960. The computation of weekly wages or earnings involved dividing by the reported number of weeks per year. In 1960 weeks are given in intervals, and we used the midpoints. Although this procedure could introduce additional variance, the results we obtain are virtually the same as for those working 50 to 52 weeks, where the interval problem matters less. In all cases, the income data refer to the year before the census, that is, 1939, 1949, and 1959. Sample sizes, for most of the analysis, are 1940 16,104; 1950 31,812; 1960 27,786.

white men only, although in some instances we broaden the criteria to include all races for the purpose of comparing our results with those for the more recent period. The decision to limit the sample to wage and salary earners was dictated by the nature of the 1940 census, which did not include information on income from self-employment. The criteria of working at least 40 weeks and receiving more than half the minimum wage were chosen to eliminate individuals whose average weekly earnings were too low to be believed. These assumptions are, conveniently, virtually identical to those used in much of the recent work on the wage structure (e.g., Juhn, Murphy, and Pierce, [1989]; Katz and Murphy, [1992]).

The wage structure can be analyzed along several dimensions among which education and labor market experience have received the most attention. Accordingly, Table II gives the ratio of weekly wages for college graduates to high-school-only graduates, the ratio of weekly wages for those with some college to high-school-only graduates, and the ratio of weekly wages of high-school-only graduates to those who stopped their education after graduating from elementary school, for 1940 to 1960. Unlike the other tables in this section, the sample covered in Table II includes all males, rather than just whites to provide exact comparability with the analysis in Katz and Murphy [1992] of the post-1963 wage structure. We have also listed ratios for 1963 and 1987. It should be noted that the ratios in Table II for 1940, 1950, and 1960 are about 10 percent larger than are those computed from a regression that holds constant various factors, including race and region. The trend over time and the comparisons with the more recent period are unaffected, however.

Although few Americans completed college in 1940 (see Table X), the ratio of weekly earnings of college to high school graduates is examined first because it is so often used for the post-1960s period. The ratio was considerable in 1940, particularly for men with fewer than 21 years of labor market experience, and its level, at about 1.6, was about equal to that achieved in 1987.⁶ There is a sharp drop-off in the ratio of weekly earnings for college to high school graduates at 21 years of experience, although only a small fraction of the college-educated sample was in that range. Measured across all experience groups (not given in the table), there

6. It should be noted that about 75 percent of all college graduates were in the ≤ 20 years experience group.

TABLE II
RATIOS OF WEEKLY WAGES FOR EDUCATIONAL GROUPS BY EXPERIENCE

College graduate/H. S. graduate Experience group	1940	1950	1960	1963	1987
1-5 years	1.742	1.357	1.528	1.538	1.880
6-10	1.728	1.369	1.522	1.397	1.704
11-15	1.777	1.461	1.565	1.584	1.635
16-20	1.652	1.421	1.602	1.462	1.689
21-25	1.408	1.449	1.627	1.516	1.722
26-30	1.475	1.466	1.580	1.467	1.661
31-35	1.393	1.367	1.641	1.390	1.533
36-40	1.357	1.189	1.398	1.682	1.568
Some college/H. S. graduate Experience group	1940	1950	1960		
1-5 years	1.195	1.029	1.128		
6-10	1.184	1.151	1.198		
11-15	1.219	1.147	1.270		
16-20	1.316	1.215	1.218		
21-25	1.164	1.276	1.192		
26-30	1.126	1.063	1.215		
31-35	0.989	1.084	1.280		
36-40	0.999	1.204	1.118		
H. S. graduate/eighth grade Experience group	1940	1950	1960		
1-5 years	1.376	1.442	1.222		
6-10	1.463	1.322	1.403		
11-15	1.377	1.299	1.326		
16-20	1.385	1.267	1.351		
21-25	1.371	1.221	1.332		
26-30	1.427	1.286	1.280		
31-35	1.354	1.250	1.243		
36-40	1.283	1.326	1.352		

Notes. The sample includes all males, 18 to 64 years old who were wage and salary earners working more than 39 weeks and earning more than one half the minimum wage on a full-time basis. Experience = (age - years of schooling - 6). High school graduates terminated their education at graduation, as did those having only eighth grade educations. Those with some college did not graduate college, but college graduates include those who continued beyond college. The numbers for all experience groups in each year have not been given because they are misleading for periods of substantial educational increase. In 1940, for example, the overall ratio of weekly earnings among high school-only graduates to those who left after the eighth grade is 1.081, a number substantially below any of the experience group averages. The reason is that high school graduates were much younger than those who left after primary school, and the overall average aggregates across age or experience groups.

Sources. 1940, 1950, 1960 PUMS. Data for 1963 and 1987 are from the March Current Population Surveys and were provided by Lawrence Katz.

was a thirteen-percentage-point decline in the premium to college over high school graduation between 1940 and 1950. The college graduation premium increased during the 1950s, but was still lower overall in 1960 than in 1940.⁷ The decline in the premium during the 1940s was concentrated among men with fewer than 21 years of work experience, for whom the decline was about 30 percentage points, while the increase in the 1950s occurred for men in all experience groups.⁸

Although there are some differences by experience group, the overall college graduation premium in 1960 is very similar to the estimate for 1963 from Katz and Murphy [1992]. The close similarity of the overall ratios in the 1960s bolsters confidence in comparisons across the entire period, 1940 to 1987. Such comparisons reveal a narrowing in the wage structure from 1940 to 1950 and a subsequent widening from the 1950s to 1987. For example, among those with 11 to 15 years of experience, the decline from 1940 to 1950 was 31.5 percentage points, the rise from 1950 to 1960 was 10.3 points, and that from 1963 to 1987 was 5.1 points. For the 6 to 10 year and 16 to 20 year experience groups, a decrease from 1940 to 1950 of between 24 to 36 percentage points was offset by an increase of almost equal magnitude from 1950 to 1987. Thus, the rise in the college to high school graduation premium during the 1980s has returned it to the level it attained on the eve of World War II.

Similar patterns are apparent in the relative wages of workers

7. Using data from published census tables on earnings by education groups, Miller [1966] found no decline in the college graduation premium between 1940 and 1950. Bartlett [1978] criticized Miller's conclusion, on the grounds that the 1940 and 1950 tables were not comparable. Using the 1970 PUMS, Bartlett constructed consistent pairwise comparisons between 1970 and published earnings tables for 1940, 1950, and 1960. She found higher returns to schooling in 1940 than in 1970, but similar returns in 1950 and 1970. From these comparisons Bartlett concluded that the returns to schooling must have fallen between 1940 and 1950, but she was unable, at the time, to substantiate the conjecture directly. Our regression results, based on the 1940 and 1950 PUMS, confirm Bartlett's conjecture.

8. The sharp decrease in the premium to college graduation with men having more than twenty years of experience may be related to a change in the quality of college education in the 1920s. Note that the same drop-off is apparent in the data for 1950, except that it occurs for men with more than 30 years of experience. Another possible explanation has to do with sample selection bias. Our sample contains only wage and salary earners, but college graduates were disproportionately self-employed. Further, college graduates were more likely to be self-employed at older ages and probably had higher incomes than the average college graduate. In the 1940 census, for example, 12 percent of the nonfarm group were self-employed, 24 percent of college graduates were self-employed, and 33 percent of college graduates 45 years old and older were. By omitting the self-employed (by necessity, since the 1940 census does not have income from self-employment), we could bias downward the returns to college among the older group.

with some college and for the high school graduate premium, although the changes are attenuated compared with those for college graduates. As in the college graduate results, there is a decline from 1940 to 1950 and a subsequent rise, but generally not large enough to revert the ratio to its previous level.

Occupational premiums were also compressed during the 1940s, as can be seen in Table III. Relative to all nonfarm workers, professionals who were wage earners lost twenty percentage points or 14 percent, while laborers gained thirteen percentage points or 21 percent. Operatives and service workers also gained somewhat compared with all nonfarm workers.

Although we do not present a separate table for the compression between experience groups, the change from 1940 to 1950 follows that for education and occupation. For high school graduates only, the ratio of the weekly earnings of those with 26–30 years of experience to those with 6–10 years was 1.75. By 1950 it was 1.39. For those with below a high school diploma, it was 1.62 in 1940 and fell to 1.30 by 1950. And for those with some college and above, it was 1.53 in 1940 but 1.42 in 1950. Further, in all cases the 1960 ratio is not much different from that achieved in 1950. Thus, the less experienced gained relative to the more experienced within all educational groups. A large literature has addressed the impact on the wage structure of the relatively small cohort that came of age in the 1950s (e.g., Easterlin [1978]; see also Freeman [1979], on subsequent changes). That literature has claimed that the decrease

TABLE III
WEEKLY WAGES RELATIVE TO ALL NONFARM WEEKLY WAGES BY OCCUPATION

	1940	1950	1960
White collar	1.256	1.177	1.192
Blue collar	0.860	0.891	0.876
Professional	1.474	1.254	1.222
Clerical	0.988	0.940	0.878
Craft	1.039	1.023	0.979
Operative	0.856	0.861	0.828
Laborer	0.630	0.750	0.736
Service	0.737	0.779	0.735

Notes. The sample includes white males, 18 to 64 years old who were wage and salary earners working more than 39 weeks and earning more than one half the minimum wage on a full-time basis. White collar includes professional, semiprofessional (in 1940), managerial, clerical, and sales workers. Blue collar includes craft, operative, laborer, and service workers.

Sources. 1940, 1950, and 1960 PUMS.

in the premium to labor market experience was primarily the result of a bidding up of the wages of the "baby bust" cohort. Yet the data we have compiled indicate that the twist in the wage structure by experience and age was part of a more extensive set of changes in the wage structure that came almost a decade earlier. Its origins, therefore, may be found in a more complex set of factors than relative cohort size.

The census evidence presented thus far pertains to changes in relative wages between groups by education, experience, and occupation. Table IV reveals another component to change, that occurring within educational, experience, and occupational groups.⁹

The difference between the log of weekly wages at the ninetieth and the tenth percentiles declined for all educational groups between 1940 and 1950 (Table IV, part A). While there was some recovery to 1960, the increase was rather modest. Further, the decrease during the 1940s within each of the educational groups was remarkably similar, with the exception of the least educated. Within each of the groups having eight or more years of schooling, the wage structure narrowed by 31 to 39 percent. That is, the difference in the log of the wage at the ninetieth and the tenth percentiles was on average 1.34 in 1940 but 0.99 in 1950, with only slight variation across educational groups.

Rather similar results are found in the data by experience group (Table IV, part B). The difference from 1940 to 1950 in the 90–10 percentile measure is 32 percent to 50 percent, with the exception of those having the least experience. As in the education results, the narrowing of the wage structure within each of the groups is seemingly independent of the level. The final panel (part C) gives similar measures for various occupational groups. All reveal a substantial decrease in within-group variance from 1940 to 1950 with modest recovery to 1960. The within-group dispersion decreased most for white-collar workers, where the decline was 32 percent. But it also decreased for craft workers (25 percent), operatives (23 percent), and laborers (8 percent). As in the previous tables showing the between-group variance, Table IV reveals not

9. Although we focus on wage differentials by education, experience, and occupation, the wage structure changed in the 1940s in other dimensions. Wage differentials between agriculture and other sectors narrowed, as did interregional variation [Wright, 1986]. Interindustry differentials declined during the early years of the Great Depression, increased in the late 1930s, and then fell sharply during the 1940s (Cullen [1956]; see also Allen [1987]). But according to Cullen [p. 361] the extent of interindustry variation around 1950 appears to have been about the same as in the late 1920s.

only that there was a decrease in the 1940s but also that the decline was from a very high level.

Finally, Table V examines the distribution of the residuals from a wage equation that includes most of the factors just examined, with the exception of occupation. The residual distribution from such a regression is an indication of the within-group variance, and thus the change in the distribution indicates alteration in the within-group variance. From 1940 to 1950 there was

TABLE IV
INTERDECILE RANGES OF THE LOG OF WEEKLY WAGES BY EDUCATIONAL,
EXPERIENCE, AND OCCUPATIONAL GROUPS

Log wage at the ninetieth percentile – Log wage at the tenth percentile					
Part A: Educational group					
	1940	1950	1960	Difference 1950–1940	Difference 1960–1950
< 8 years	1.347	1.211	1.166	–0.136	–0.045
8	1.378	0.988	1.034	–0.390	0.046
9–11	1.369	0.976	1.073	–0.393	0.097
12	1.312	0.952	1.136	–0.361	0.184
13–15	1.460	1.146	1.263	–0.314	0.117
≥ 16	1.540	1.207	1.312	–0.333	0.105
Part B: Experience group					
	1940	1950	1960	Difference 1950–1940	Difference 1960–1950
1–5 years	1.253	1.246	1.429	–0.007	0.183
6–10	1.347	1.032	1.177	–0.315	0.145
11–15	1.347	1.004	1.088	–0.343	0.083
16–20	1.426	0.921	1.067	–0.504	0.146
21–25	1.291	0.978	1.058	–0.313	0.080
26–30	1.427	0.996	1.088	–0.431	0.092
31–35	1.416	1.019	1.109	–0.398	0.091
36–40	1.455	1.041	1.109	–0.414	0.068
Part C: Occupational group					
	White	Blue collar			
	Collar	Craft	Operative	Laborer	All
1940	1.465	1.105	1.090	1.138	1.236
1950	1.146	0.856	0.881	1.059	1.017
1960	1.238	0.913	0.979	1.028	0.889

Notes. The sample includes white males, 18 to 64 years old who were wage and salary earners working more than 39 weeks and earning more than one half the minimum wage on a full-time basis. White collar includes professionals, semiprofessionals (in 1940), managers, and clerical and sales workers. Blue collar includes craft, operative, laborer, and service workers.

Sources. 1940, 1950, and 1960 PUMS.

TABLE V
CHANGE IN THE DISTRIBUTION OF RESIDUALS FROM A LOG WAGE EQUATION

Difference between deciles	1940	1950	1960
90-10	1.115	0.906	0.953
90-50	0.513	0.428	0.446
50-10	0.602	0.478	0.507
σ^2_{μ}	0.197	0.177	0.171

Notes. The sample includes white males, 18 to 64 years old, who were wage and salary earners working more than 39 weeks and earning more than one half the minimum wage on a full-time basis. The residuals are derived from wage equations that regress on the log of weekly wages: experience (= age - years of school - 6), experience squared, years of school, schooling squared, the interaction of experience and schooling, regional dummies, marital status dummies, head of household dummy, veteran status (for 1950 and 1960), and city or urban or SMSA area. σ^2_{μ} is the variance of the residual.

Sources. 1940, 1950, and 1960 PUMS.

an increase at the lower end of the residual distribution and a decrease at the upper end, and the change, once again, is an almost mirror-image of that from 1963 to 1987 [Katz and Murphy, 1992]. Both the residual variance and the three decile differences given in Table V indicate a narrowing in the distribution of the residual. Thus, the decrease in the within-group variance, as seen in Table IV, is not an artifact of failing to control simultaneously for the effects of education, experience, region, and so on, or from focusing on particular centiles of the wage distribution. Rather, there was a decline in the within-group variance that cannot be explained entirely by the factors considered thus far.

B. Decomposing Change in the Wage Structure

A simple variance decomposition, using the regressions underlying Table V, indicates that about 30 percent of the decrease in the variance in the log of weekly wages was due to a decrease in the variance of the residuals and 70 percent to a decrease in the variance of the quantities and market prices of observed covariates.¹⁰ But a simple variance decomposition does not tell us much about how changes in prices, quantities, and the residual distribu-

10. The decrease in the variance in the residual accounts for 28.8 percent of the total change in the variance of (the log of) wages. The variance decomposition also indicates that changes in prices, not quantities, are responsible for most of the compression in wages. If only experience and schooling are included in the regression, thus excluding region, marital status, city residence, and so on, 42.4 percent of the decrease in the variance in weekly wages was due to a decrease in the residual variance. We present that figure, as well, because the regressors are similar to those in Juhn, Murphy, and Pierce [1989].

tions alter the entire distribution of wages. To further investigate change in the wage structure over time, we follow Juhn, Murphy, and Pierce [1989] and construct a decomposition based on the standard wage equation:

$$(1) \quad y_{it} = X_{it}\hat{\beta}_t + \mu_{it},$$

where y_{it} = the log of weekly wages for individual i in year t , X_{it} = individual characteristics that affect weekly wages, and β_{it} = the implicit prices or values for these characteristics. The error term in equation (1) can be thought of as consisting of two components and taking the form,

$$(2) \quad \mu_{it} = F_t^{-1}(\phi_{it}|X_{it}).$$

That is, individual i occupies some position or percentile in the distribution of residuals for year t , ϕ_{it} , and each year has a particular inverse cumulative distribution of residuals, $F_t^{-1}(\cdot | X_{it})$. Framed in this manner, the percentile occupied by each individual and the distribution function for each year can be varied independently. Changes in the wage structure can then be decomposed into three factors: changes in the quantities of skills (the X 's), changes in the prices of skills and other variables (the $\hat{\beta}$'s), and changes in unobservable factors that get swept into changes in the distribution of residuals.

The virtue of the technique is that it reveals how the full distribution of wages changes as quantities and prices are successively changed. A complication is that there is no unique decomposition, and the procedure may be sensitive to the order of the changes as well as to the base-year weights employed. We compute the technique in several ways to test the robustness of the findings.

Intuitively, one computes the full distribution of wages for some base year, say 1940, and from this distribution various summary statistics, such as the difference in the log of wages at the ninetieth and the tenth percentiles. A hypothetical experiment is then conducted which consists of changing the characteristics (the X 's) of individuals. The full distribution of the resulting wages is then computed and from this, the summary statistics. The difference between, say, the 90–10 measure for the actual distribution and that in the hypothetical world of changed quantities, is allotted to movements in quantities. The next step is to change both prices and quantities and allocate the incremental change to prices. Finally, the total change in the 90–10 measure minus that due to prices and quantities is assigned to changes in the residual

distribution. Note that both the base year and the order of the decomposition are arbitrary.

Formally, consider the log wage for an individual in 1950:

$$(3) \quad y_{1950} = X_{50}\hat{\beta}_{50} + F_{50}^{-1}(\phi_{50}|X_{50}),$$

where the individual i subscripts have been dropped for convenience. A predicted log wage, y_1 , is then computed for each individual in 1950 with coefficients from a 1940 regression and a residual that takes the individual's position (ϕ) from the residual distribution of 1950 but assigns the corresponding residual value from the 1940 distribution:

$$(4) \quad y_1 = X_{50}\hat{\beta}_{40} + F_{40}^{-1}(\phi_{50}|X_{50}).$$

The only difference between the hypothetical y_1 of equation (4) and the actual y_{1940} , given by equation (5),

$$(5) \quad y_{1940} = X_{40}\hat{\beta}_{40} + F_{40}^{-1}(\phi_{40}|X_{40}),$$

concerns the characteristics (the X 's). Next, change both the X 's and the β 's to get

$$(6) \quad y_2 = X_{50}\hat{\beta}_{50} + F_{40}^{-1}(\phi_{50}|X_{50}).$$

In the case being considered, y_2 will have a more compressed distribution than y_1 , and the increment to the compression is allocated to the factor that has changed between equations (4) and (6), the β 's. That is, comparing the distribution of y_2 with that of y_{1940} reveals how both prices and quantities affect the distribution of wages. The final step is to subtract the total of the changes, that is, the particular summary statistic of the distribution of y_2 , from the total change over the two years. Comparing y_2 with y_{1950} shows that the difference is due solely to the change in the residual distribution.

Table VI shows the decrease in the wage spread from 1940 to 1950 using three summary measures of the wage distribution, calculated from data in Table I. Using the 90–10 percentile measure, the change was a decrease of 0.354, that is, a 35.4 percent decrease in spread. There was a 20.2 percent decrease in the 90–50 percentile measure, that is, from the median to the high end of the distribution, and a 15.3 percent decrease in the 50–10 percentile measure, that is, from the median to the low end.

The decomposition reveals the importance of changes in prices over the two periods, although quantity changes are almost as important using 1950 base weights. Price changes, however, overwhelm quantity changes using 1940 as the base year. The

TABLE VI
DECOMPOSITION OF THE CHANGE IN THE WAGE STRUCTURE, 1940 TO 1950

Using 1950 as the base year				
Difference in percentiles	Total change log (wages) ^a	Due to change in		
		Observed quantities	Observed prices	Distribution of residuals
90-10	-0.354	-0.172	-0.167	-0.016
90-50	-0.202	-0.088	-0.077	-0.036
50-10	-0.153	-0.083	-0.090	+0.020

Using 1940 as the base year				
Difference in percentiles	Total change log (wages) ^a	Due to change in		
		Observed quantities	Observed prices	Distribution of residuals
90-10	-0.354	+0.007	-0.143	-0.219
90-50	-0.202	-0.014	-0.074	-0.114
50-10	-0.153	+0.021	-0.069	-0.105

a. Total change for the 90-10 row, for example, is the (log wage at the ninetieth percentile - log wage at the tenth percentile)₁₉₅₀ - (log wage at the ninetieth percentile - log wage at the tenth percentile)₁₉₄₀.

Notes. The decomposition is described in the text. The wage equation (1) was estimated for 1940 and 1950 on the sample of white male wage and salary earners, 18 to 64 years old, whose (full-time) weekly wage and salary earnings were more than one half the current minimum wage. The variables included are experience (= age - years of school - 6), experience squared, years of school, schooling squared, an interaction of schooling and experience, dummy variables for high school and college graduation, for region, for urban residence, for marital status, for household headship, and for foreign birth.

Sources. 1940, 1950, and 1960 PUMS. See text for a description of the decomposition procedure.

choice of base year greatly affects the importance of changes in the distribution of residuals relative to changes in both prices and quantities. When 1950 is the base year, the residual distribution has virtually no impact on the total change, while it accounts for about 60 percent when 1940 is the base year. Although the decomposition does not yield unambiguous results on the relative importance of the residual distribution, it provides a clear statement on the role of the prices of skills. The returns to schooling plummeted during the 1940s and that factor, together with the other price changes, were responsible for about 45 percent of the compression in the wage structure. What ultimately caused the price of skill to decline is the subject of the next section.

II. EXPLAINING THE GREAT COMPRESSION

The compression in the wage structure from 1940 to 1950 may have been caused by factors unique to the World War II period or,

alternatively, may have been a continuation of secular forces disrupted during or accelerated by the Great Depression. Although the pre-1940 evidence is slender, a decline in the skill differential is apparent between the wages of manufacturing operatives and craftsmen. Similar findings for the earnings of the upper portion of the income differential also suggest a pre-1940 decrease in inequality. These data, therefore, imply that the compression of the 1940s was a continuation of secular forces. But the decline before 1940 was of a considerably smaller magnitude than that during the 1940s, and the Great Compression was not, in any way, a simple extrapolation of what came before.

Further, we present below new data on white-collar earnings that reveal, when compared with wages for unskilled labor, an absence of a pre-1940 compression in the skill differential. The series even suggest a widening in the returns to skills, such as education, during the 1920s. Therefore, even though the ratio of skilled manufacturing worker wages to unskilled manufacturing worker wages may have declined prior to 1940, the returns to education may not have begun to narrow until the 1940s.

Our white-collar wage series reveals another factor of importance to this study. Because unemployment during the Great Depression was disproportionately borne by the least educated and the lesser skilled, one might be suspicious of the earnings data for 1939 (from the 1940 census). But the new series demonstrates that 1939 was not anomalous.¹¹ Although inequality widened at the beginning of the 1930s, it narrowed after 1933, probably because of the minimum wages of the National Industrial Recovery Act (NIRA). Most series we have uncovered show little overall change from 1929 to 1939, although the wage structure changed within the decade.

Because the wage structure in 1939 was not anomalous, we are led, therefore, to factors unique to the World War II period to explain much of the initial compression. The National War Labor Board may have engineered a large portion of the narrowing during the first half of the decade. Evidence from a host of industry wage spanning the late 1930s to the early 1950s, reveals that studies,

11. We emphasize that we have evidence only on differences in wages between white-collar and various blue-collar occupations for the 1920s and 1930s. The within variance for both groups may have widened in the early 1930s and may have remained so until the 1940s.

compression during the 1940s was not confined to the period before V-J Day. The introduction and expansion of the minimum wage had some impact, but cannot explain the compression in the wage structure at the upper end of the distribution. Further, although the wage structure widened somewhat after 1950, it remained considerably more compressed than before the war. To explain the persistence of the command economy wage structure, we return to educational changes which increased the relative supply of more skilled workers and to other influences which buoyed the relative demand for the lesser skilled.

A. Long-Run Factors Affecting the Wage Structure

Historical statistics have not been kind to scholars interested in the wage structure. The 1940 census was the first to inquire into the earnings and education of Americans, and before that date no comprehensive data exist that include information on education and earnings. Information is available, however, to produce occupational wage ratios across long periods of time. The numerator of the pay ratio has most often been the wage of skilled artisans, typically in the building trades but also skilled craftsmen in manufacturing. The denominator has often been that of "common laborers," but also janitors, helpers, maintenance workers, hand truckers, and manufacturing operatives, particularly in the twentieth century.

Williamson and Lindert ([1980]; hereafter W&L) have linked various series to produce a composite series of pay ratios between artisans and laborers extending from the early nineteenth to the mid-twentieth century. Portions of their linked series for the twentieth century are given in Table VII, together with a series (National Industrial Conference Board, NICB, column 1) that underlies their index, one due to Ober [1948] and Miller [1966], and another to the BLS (column 3). The data show a slow secular increase in the pay ratio of the skilled to the unskilled until about 1925, with the exception of an important decrease during World War I. From 1925 to 1948 the ratio declined by 10 percent. But, measured in logs, about two thirds of the decline occurred during the 1940s. Widely cited series constructed by Ober and Miller, from data for the building trades, show a more continuous narrowing from 1907 to 1940, and a much steeper decline from the late 1940s to the early 1950s (Table VII; see also Keat [1960]). The discrepan-

TABLE VII
PAY RATIOS OF SKILLED TO UNSKILLED WORKERS, 1907–1960

	(1)	(2)	(3)	(4)		(5)	(6)
	NICB	W&L	Ober &	Railroad		Railroad	NYS
	(hourly)	Index	Miller	clerks/		machinists/	clerks/
			[BLS]	laborers		laborers	unskilled
				(monthly)	(hourly)	(hourly)	(weekly)
1907		1.849	2.05				
1909		1.909					
1917		1.876					
1918		1.722	1.75				
1922	1.408	1.943		1.57	1.57	2.03	
1923	1.397			1.53	1.57	1.95	1.89
1924	1.406			1.58	1.58	1.92	1.95
1925	1.415	1.952		1.60	1.59	1.94	1.94
1926	1.414			1.60	1.60	1.95	1.96
1927	1.393			1.64	1.63	1.97	1.99
1928	1.390			1.69	1.68	1.96	1.96
1929	1.374			1.69	1.68	2.04	1.98
1930	1.387	1.922		1.79	1.71	2.06	2.25
1931	1.378		1.80	1.93	1.74	2.06	2.41
1932	1.398			2.14	1.77	2.06	2.91
1933	1.372			2.06	1.81	2.09	2.78
1934	1.342			1.98	1.81	2.08	2.60
1935	1.343	1.880		1.97	1.82	2.10	2.30
1936	1.375			1.85	1.79	2.09	2.13
1937	1.363			1.81	1.75	2.03	2.00
1938	1.369		1.65	1.81	1.72	1.95	2.11
1939	1.360	1.888		1.75	1.71	1.95	2.01
1940	1.354			1.73	1.70	1.95	1.89
1941	1.340			1.65	1.63	1.89	1.77
1942	1.349			1.57	1.54	1.81	1.68
1943	1.363			1.44	1.43	1.69	1.49
1944	1.376			1.39	1.42	1.70	
1945	1.361			1.39	1.41	1.70	
1946	1.300		1.55	1.38	1.35	1.55	
1947	1.289			1.35	1.33	1.52	
1948	1.277	1.773		1.31	1.29	1.46	
1949				1.32	1.28	1.42	
1950			[1.580]	1.34	1.27	1.41	
1951				1.32	1.25	1.37	
1952			1.37	1.31	1.23	1.34	
1960			[1.603]				

Notes. The Williamson and Lindert (W&L) index, column (2), splices together several ratios of skilled to unskilled wages. Primary among these is that from the National Industrial Conference Board (NICB), also given here in column (1) for 1922 to 1948. The column (1) NICB data are the ratio of the hourly wages of skilled and semiskilled manufacturing production workers to those for unskilled manufacturing production workers. Note that the Williamson and Lindert numbers are an index, and thus do not give the actual percentages by which skilled workers earned more than the unskilled. The BLS series is the ratio of the hourly wage rate for mechanics, electricians, and carpenters in six cities to that for janitors and custodians. The dates for the BLS series are 1950/1951 and 1960/1961. The Ober-Miller series is the ratio of hourly wages of skilled labor to that for unskilled labor in various manufacturing industries in urban areas. The dates for the Ober-Miller series are 1907, 1918/1919, 1931/1932, 1937/1940, 1945/1947, and 1952/1953. The railroad data are for clerks, machinists, and laborers working on class-I steam railroads. For the hourly earnings, total compensation was divided by the total number of hours worked; for the monthly earnings, total compensation was divided by the average number of workers in the middle of the month, which in turn was divided by twelve. The New York State clerical data are the weekly wages of male clerks working in New York State factories. The denominator is the unskilled NICB series for weekly wages.

Sources. U. S. Bureau of the Census [1975], series D-839 and D-842 for column (1). Williamson and Lindert [1980, pp. 307–08] for columns (2) and (3). Column (2) is their series (5); column (3) is their series (6) and (7), for the BLS. U. S. Interstate Commerce Commission [1922 to 1952] for columns (4) and (5). U. S. Bureau of the Census [1975] series D-841 and U. S. Department of Labor, *Monthly Labor Review* [1928 to 1945] for column (6). Goldin and Margo [1991] details the sources and the underlying series of columns (4), (5), and (6).

cies between the W&L linked and the Ober and Miller series can partly be traced to differences in the underlying data.¹²

Further evidence that the narrowing of the wage distribution began before 1940 but accelerated after is found in the works of Kuznets [1953] and Goldsmith [1967] on the income distribution. Both studies use data from the income tax. The Kuznets series for the share of income received by the top 5 percent shows a decrease from about 30 percent in the 1920s to 28 percent in 1939. It falls to 26 percent by 1941 and plummets to just below 20 percent in 1946. The Goldsmith measure indicates that a somewhat larger proportion of the decline occurred in the 1930s.

If the secular forces that operated to narrow skill differentials were those that increased schooling, then neither of the wage series just examined is very useful, since both are concerned with skill differentials between blue-collar occupations. Nor are the Kuznets and Goldsmith figures very illuminating since they concern just the very top of the income distribution. It is natural to ask whether pay ratios for white-collar workers, such as professionals and clerical workers, show evidence of decline during the first half of the twentieth century.

Existing evidence for white-collar workers presents a somewhat confusing picture of inequality changes over long periods of time. The data are often for public-sector employees (e.g., teachers), or for self-employed professionals (e.g., doctors, lawyers) whose hours cannot easily be held constant over periods of economic downturn. The ratio of teachers' earnings to those of unskilled nonfarm laborers decreased substantially in the 1930s, narrowed slightly in the 1940s, but widened since. Data for

12. The W&L linked series, from 1921 to 1948, comes from data collected by the National Industrial Conference Board [Beney, 1936; U. S. Bureau of the Census, 1975]. The NICB distinguished between two groups: skilled and semiskilled manufacturing workers, and unskilled manufacturing workers. Unskilled workers were in jobs requiring no previous work experience, while the semiskilled and skilled (the numerator) were workers in all other positions. Clerical, sales, executive, and managerial personnel were excluded. Thus, the NICB data yield, in effect, pay ratios between craftsmen and operatives in manufacturing. Evidence from the 1940 and 1950 public use census samples suggests little change in the ratio of craft to operative wages between 1940 and 1950, consistent with the NICB data. In simulations of their general equilibrium model for the period 1929 to 1948, Williamson and Lindert substituted a pay ratio for 1951 for the 1948 NICB datum, believing that the 1948 ratio was too high. The 1951 pay ratio, constructed from the BLS Area Wage Surveys, refers to mechanics, electricians, and carpenters in the building trades for the skilled group, and janitors and custodians, for the unskilled. There is no empirical basis for substituting the 1951 for the 1948 ratio. The two data sources do not overlap and, in any event, cover entirely different occupations. The only rationale could be that the Ober and Miller series shows a greater decrease than the NICB series.

physicians relative to those for manufacturing operatives reveal an upward, not a downward, trend from the early twentieth century, with some narrowing during the Great Depression. Associate professors' salaries show a clear long-run relative downward trend, with the greatest compression during the 1930s.¹³ A major problem with the available white-collar pay ratios is that the data on professional incomes are rarely adjusted for cyclical variations in hours or weeks worked, while the data for unskilled labor (the denominator) are. When ratios are adjusted for such cyclical variations, the decline in relative professional pay in the 1930s, while present, is much smaller.

B. New Evidence on the Skill Differential and White-Collar Earnings, 1920s to 1950s

Evidence on the earnings of white-collar workers is required to assess whether the compression in the wage structure we observe in the 1940s was a continuation of a longer-term secular movement or a product of more recent vintage. Because white-collar workers reap more of the returns to education than do those who are unskilled workers in manufacturing, we have constructed two series for clerical-worker wages, given as ratios to unskilled workers in Table VII. (These series are described in detail in Goldin and Margo [1991].) That in column (4) is the ratio of the hourly (or monthly) wage of clerks hired by class-I steam railroads to the hourly (or monthly) wage of laborers, also hired by these railroads. That in column (6) is the ratio of the weekly earnings of male clerks working in factories in New York State to the weekly earnings of unskilled manufacturing workers. All series move together, although change is greatest for the ratio using NYS clerks, which is unadjusted for hours. The three are graphed in Figure II.

The series on clerk to unskilled wages show some gain during the 1920s, but the interesting changes appear as the depression quickly deepened after 1929. Clerical wages rose immediately relative to unskilled wages. From 1929 to about 1934 the increase in the ratio was 8 percent for railroad workers on an hourly basis, 26 percent on a monthly basis, and 31 percent for the NYS clerical workers on a weekly basis. There is considerable evidence, therefore, that on an hourly basis the more educated gained relative to

13. The teacher, physician, and associate professor series are reproduced in Williamson and Lindert [1980] from U. S. Bureau of the Census [1975].

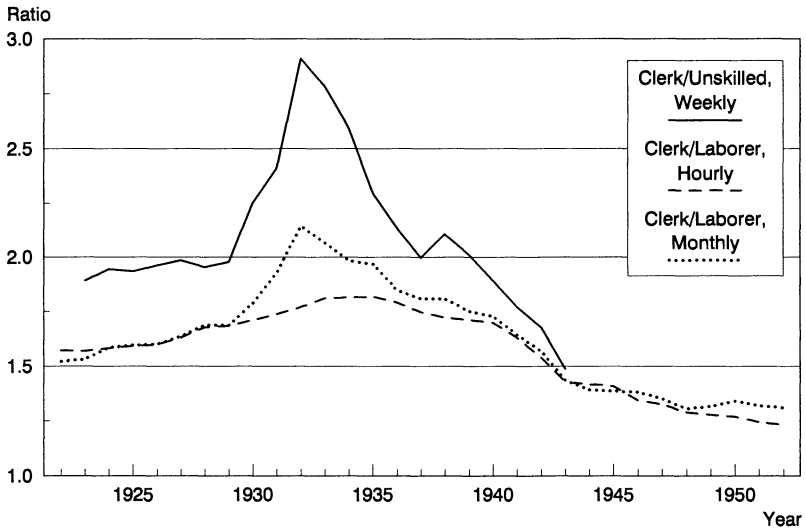


FIGURE II

Clerk Wages Relative to Unskilled and Laborer Wages, Males: 1922 to 1952

Source. Table VII, columns (4) and (6).

Notes. The weekly series for clerks is for male office workers in New York State factories, and the corresponding series for the unskilled refers to production workers in 25 industries (from National Industrial Conference Board data). The hourly and monthly series are for workers on class-I steam railroads.

the less educated during the early years of the Great Depression. And the educated gained far more on a weekly basis, as their hours declined less than did those for the employed lesser-skilled worker. Hours of laborers decreased by 26 percent and those of machinists by 30 percent, while those of clerks decreased by only 5 percent from 1929 to 1932 or 1933 (see also Bernanke [1986]; Goldin and Margo [1991] contains further details on hours).

But the relative gains experienced by clerical workers were short-lived. Some time between 1933 and 1934, depending on the series, the gains they made were eroded, and the likely reason was the setting of minimum wages and maximum hours by the National Industrial Recovery Act (NIRA), passed June 1933 (see Weinstein [1980] on the NIRA). But the NIRA itself was short-lived, and there is no apparent increase in the ratio after the NIRA was declared unconstitutional in May 1935. Quite possibly economic recovery reversed the immediate impact of the depression on the skill differential.

C. Was 1939 Anomalous?

Prior to the construction of the series just described, there was ample reason to believe that wages of the less-educated in the late 1930s had been dampened by a “reserve army” of private-sector unemployed. Data from the 1940 census sample reveal that unemployment rates were considerably higher among less-educated workers, and were U-shaped with respect to labor market experience.¹⁴ According to some scholars, the availability of large pools of unemployed workers, particularly less-skilled and less-educated labor, put downward pressure on wage rates in the late 1930s [Baily, 1983]. Because the unemployed were disproportionately less-educated, the “reserve army” effect arguably could have widened the wage distribution in 1939.

But by 1939, the year to which most 1940 census data refer, the skill ratios in Table VII, columns (4) and (6) had returned to their approximate values in 1929, despite the fact that they had widened, often significantly, during the depression. That is, 1939 does not appear anomalous in terms of the premium to educated labor. Nor, in fact, does it appear anomalous in terms of the premium to skilled labor, as in the case of railroad machinists whose hourly wages also rose relative to railroad laborers during the early years of the depression. We should emphasize that our view concerning the rebound in the relative wages of the unskilled during the last half of the 1930s is based on two new series for educated labor. Further evidence may lead us to revise this

14. In the 1940 sample of white males, 18 to 64 years old, unemployment rates that include persons on work relief arrayed by educational attainment yield:

Years of schooling	Percentage unemployed (or on work relief)
<8	21.0%
8	17.6
9–11	17.6
12	15.3
13–15	10.1
≥ 16	6.2

A U-shaped pattern of unemployment with respect to experience (age – years of school – 6) is revealed by linear probability regressions in which the dependent variable takes the value of one if the person was unemployed (or on work relief), zero otherwise. In addition to experience, the regressions include dummy variables for region, urban status, marital status, and household headship. Separate regressions were estimated for each educational group.

conclusion. But there are currently no available wage series based on large samples that indicate the wage structure was anomalous in 1939 in comparison to the late 1920s. If anything, the previously used series (based on the NICB data) indicate virtually no change at all during the Great Depression.

In sum, the premium to skill, particularly to that associated with schooling, does not appear to have declined prior to the 1940s. In fact, there is some evidence that it widened during the 1920s and fluctuated, with no permanent change, during the depression. The skill premiums for both educated and trained labor returned to their predepression levels by 1939. It should also be noted that the clerical data reveal that the ratio of hourly earnings of clerks to laborers continued to decline even after 1945, reaching a level just above 1.2 in 1952.¹⁵ Because we can be fairly confident that the wage structure observed using the 1940 census had not been stretched considerably during the depression, we can venture to ask what caused it to be compressed during the 1940s.

D. World War II and the National War Labor Board

Any explanation for the Great Compression of the 1940s must closely examine the impact of World War II and the command economy that accompanied it. Even though the war probably increased the demand for less skilled relative to more skilled and educated labor, firms could not simply increase wages to attract labor. The National War Labor Board (NWLB), established in January 1942 and dissolved in December 1945, was responsible for approving all wage changes after the Stabilization Act (also known as the Price Control Act) of October 1942 made any wage increase illegal without its approval.¹⁶ During its brief lifetime, it and its regional agencies processed 463,000 applications for wage increases and ruled on 70,000 cases of possible violation of the Stabilization Act [U. S. Department of Labor, 1949].

Given its limited staff, the NWLB made decisions based on rules of thumb concerning wage increases in war and nonwar

15. BLS survey data also show smaller increases in nominal hourly earnings of white-collar workers between 1941 and 1944 than of manufacturing operatives or common labor [U. S. Department of Labor, 1944, p. 1048]. Increases in weekly earnings in manufacturing were greater still because the percentage increase in weekly hours was more in that sector than elsewhere, and because a disproportionate share of the increase in manufacturing hours was compensated at overtime rates.

16. Although the NWLB was dissolved in December 1945, wage and price controls were maintained, to a degree, for some time after. The post-V-J Day controls, however, were not as successful as the wartime controls had been (see Dunlop [1949]).

industries. One of these, known as the “Little Steel Formula,” was decided in July 1942, before the NWLB obtained expanded powers under the Stabilization Act. Steel employees were granted a small wage increase on equity grounds, but a larger—15 percent—increase to adjust for cost of living changes from January 1941 to May 1942. With “Little Steel,” any wage could rise by 15 percent over its January 1941 level. Although “Little Steel” by itself could not affect the wage structure, it held the line, and thus other rules of thumb could and did alter the wage structure. Wages could be increased to 40 cents per hour without the NWLB’s approval and up to 50 cents per hour with approval of the Regional NWLB, following President Roosevelt’s directive to increase substandard wages [Rockoff, 1986, p. 119]. Ranges of wages, called “brackets,” were established for each occupation in each region, and wages could be raised to the lower end within any bracket. The intent of the brackets was to reduce interplant differences in wages.¹⁷ The NWLB also allowed increases that eliminated intraplant differences in wages by occupation. Finally, the NWLB, following its mandate to assist in “the effective prosecution of the war,” allowed certain wage increases in war-related industries (see Rockoff [1986] for a more complete discussion). Most of the criteria for wage increases used by the NWLB served to compress wages across and within occupations, industries, and plants. To be sure, firms found ways to circumvent wage stabilization—the payment of benefits rather than wages and the upgrading of workers into better-paying jobs, were two. But exceptions to controls, it was often noted, were more frequently granted to employees receiving “substandard” wages.¹⁸ Thus, the NWLB may have served to compress the wage structure significantly.

Industry studies done by the Bureau of Labor Statistics from the late 1930s to the early 1950s are used here to assess precisely when the wage structure was compressed during the 1940s. We had inferred a narrowing of the wage structure during the 1940s from census data on earnings in 1939 and 1949, but such data cannot reveal the precise role of World War II. Although the war

17. See Dunlop [1950] on wage stabilization and the NWLB and on brackets, in particular. In each locality a “sound and tested” minimum wage rate was established for principal occupations in every industry. Wages that were more than a specified percentage below the mean were allowed to increase. It appears that the “brackets” rule of thumb could have had a major effect on the within variance of wages by occupation, education, and job experience.

18. See Ober [1948, p. 132] on the general issue. Data from a special census survey in 1944 reveal that occupational shifts were greater among less-skilled and less-educated groups [Wool and Pearlman, 1947, p. 141].

could have served to narrow the wage structure through the operation of the National War Labor Board, a subsequent widening could have accompanied the dismantling of controls. Alternatively, the wage structure could have changed in both periods or in the post-World War II period but not during the war. The fact that the wage structure did not quickly return to its pre-World War II configuration for more than fifteen years after the war, is *prima facie* evidence that various forces served to reinforce the impact of the command economy. The existence of a vast number of BLS reports enables a fuller understanding of the role of World War II and the NWLB. The reports contain wage distributions by industry, and often within industry by region, sex, and skill. Many were background reports for legislation on the minimum wage, which was increased four times from its inception at 25 cents with the Fair Labor Standards Act of October 1938 to January 1950, when it was increased to 75 cents.¹⁹ It was increased to 30 cents in October 1939 and again in October 1945, just after the war, to 40 cents.

The BLS surveyed more than 70 (at about the three-SIC-digit level) industries from the late 1930s to the early 1950s, but only fifteen industries contain comparable information for two periods during the time interval. These data are summarized in Table VIII (see Goldin and Margo [1991], for the actual numbers and for more complete documentation). For only three industries (cotton textiles, fertilizer, and foundries) are there wage distributions for the prewar (1936 to 1941), war (1942 to 1946), and postwar (1947 to 1952) periods. For all other cases there are two observations, most covering the prewar to war or war to postwar periods. From these we can make inferences about the timing of the wage compression and its causes. The dating of the prewar, war, and postwar periods

19. Minimum wage legislation from its inception to January 1950 can be summarized as follows:

Effective date of minimum wage change	Nominal minimum wage	Percent of employees covered	Minimum wage/average hourly wage in manufacturing	
			Before	After
10/24/38	\$0.25	43.4%		0.403
10/24/39	0.30	47.1	0.398	0.478
10/24/45	0.40	55.4	0.295	0.394
1/25/50	0.75	53.4	0.278	0.521

Source. Ehrenberg and Smith [1991, p. 83].

TABLE VIII
WAGE COMPRESSION IN THE 1940S: DIFFERENCES IN THE LOG (HOURLY) WAGE AT VARIOUS DECILES DURING THE PREWAR, WAR, AND POSTWAR PERIODS

Industry	90-10			50-10			90-50		
	Prewar to war	War to postwar	Prewar to postwar	Prewar to war	War to postwar	Prewar to postwar	Prewar to war	War to postwar	Prewar to postwar
Chemicals, industrial									
U.S.		-			+			-	
Mid-Atlantic		+			+			-	
New England		-			-			-	
Cigars	+			+			+		
Cotton textiles	-	+	-	-	+	-	-	-	-
Dyeing and finishing, cloth	-			-			-		
Fertilizer	-	-	-	-	-	-	-	-	-
Foundries	-	-	-	-	-	-	-	-	-
Grain-mill products		-	-		-	-		-	-
Flour-milling		-	-		-	-		-	-
Iron and steel		-	-		-	-		-	-
Lumber									
U.S.	+			-			+		
North	-			-			-		
South	+			+			-		
West	-			-			+		
Men's neckwear			+			+			+
Sawmills, South ^a		+			+			+	
Shipyards, private	-			-			-		
Soap									
Steel, fabricated structural			-			-			-
U.S.	n.a.							n.a.	
Southeast	-				-			-	
Woolen and worsted mills		-			+			-	

a. Small positive change for 10/46 to Fall/49 and large negative change for 10/46 to 3/50, after passage of the 75 cents minimum wage.

Notes. Prewar years are 1936 to 1941; war years are 1942 to 1946; postwar years are 1947 to 1952. A - sign means a decrease in the difference of the log wage for the deciles given and a + means an increase.

Source. U. S. Department of Labor, *Monthly Labor Review* [1938 to 1952]. See Goldin and Margo [1991] which includes the values of the log (hourly) wage at the given deciles, the dates of the surveys, median wages for the industries, percentage female, total employment, and the exact citations.

is meant to capture, first, the wage distribution before the National War Labor Board, next the distribution after its rules took effect, and later still, the wage structure when firms had adequate time to respond to the dismantling of controls.

The main finding is that wage compression in the upper range of the distribution (measured by the difference in the log of the wage at the ninetieth and fiftieth deciles) occurred during the war and continued long after wage controls ended, a finding from the clerical data as well (see Table VII). Compression in the lower portion of the distribution (from the tenth decile to the median) was generally strong in the prewar to war period but less strong in the war to postwar period, despite the continued rise of the minimum wage. On net, the more inclusive measure of the wage distribution (the difference in the log of wages at the ninetieth and tenth deciles) mirrors the large changes in the upper tail, and the notion that the wage structure experienced compression during both the war and postwar periods is upheld.

The exceptions to the general trends just mentioned are instructive. Industries that were female-intensive (cigars, men's neckwear, woolen and worsted mills) comprise an obvious exception. Rather than experiencing a narrowing of the wage structure from the prewar to the wartime or postwar periods, their distributions actually widened. Wages in these industries, like so many that are female-intensive, were extremely low and contained little dispersion prior to the war. Wartime demand increased women's wages relative to men's and led to an increase, not a decrease, in wage dispersion among workers in female-intensive industries. Also of interest are the southern industries, among which sawmills and the lumber industry experienced general wage expansion, particularly in the 50–10 range. Because so many workers in these industries were at or near the minimum wage in particular years, change in the lower portion of the wage structure is highly dependent in the date of the survey. It should be noted, however, that there was a narrowing in the wage structure for southern lumber in the 90–50 range from the war to postwar periods, and thus it, too, seems to follow the overall trend.

Explanations for the wage compression that rely on the activities and impact of the NWLB would imply that war-related industries had less compression from the prewar to war periods and that low-wage industries had more compression, particularly if the initial wage distribution was highly dispersed. The BLS data do not enable a test of the hypothesis regarding war-related sectors.

Although workers in private shipyards did experience wage compression, the industry grew by such an enormous factor that the character of the work force may have been completely altered. Chemicals, rubber, fabricated metals, and transportation machinery—the main wartime industries—are not represented in the prewar sample.²⁰ The hypothesis concerning the lower wage industries, particularly with regard to those having the most dispersion, does seem correct. The greatest declines were in textiles, a low-wage industry, and foundries, in which a large proportion of the work force was below the 55 cents hourly wage eventually allowed by the NWLB.

The evidence from the BLS reports, therefore, reinforces the notion that World War II and, possibly, the NWLB had large effects on the wage structure but that compression continued long after the war. Further, wage compression during the war now appears more complicated than suggested by a simple application of the rules of thumb used by the NWLB, which allowed wage increases at the lower end of the distribution. The upper tail of the distribution, even in relatively high wage and nonwar related industries, also narrowed considerably during the war. This fact, in combination with the relative stability in the wage structure from 1949 to 1959, bolsters the notion that market forces increased the demand for less-skilled manufacturing workers on a continuing basis from the 1940s to the early 1960s and served to keep in place a wage structure that was initially necessitated by the wartime economy and mandated by the command economy. This conclusion leads us to consider the roles of a changing demand for educated labor given fixed technology and the role of a changing supply of educated labor.

E. Changes in the Demand for and Supply of Educated Workers, 1940 to 1980

We first analyze the demand for (white male) educated workers by holding constant, at its 1960 value, the proportion of the (white male) labor force, by industry, that achieved a particular level of schooling and varying employment by industry over the years (see Freeman [1975] for a discussion of the demand index computed). The findings are summarized in Table IX. Not surprisingly, the demand for white male workers with more than twelve years of schooling increased faster than that for white male

20. Wage structure data exist for foundries in all three periods, although different types of foundries are included in each. Changes in foundry wage structures do not appear different from those in nonwar related industries. That is, there does not seem to have been more growth in the upper tail of the distribution.

TABLE IX
THE DEMAND FOR EDUCATED LABOR: 1940 TO 1980

Decade	Average annual percentage change in the demand for white males with		
	(1) Fewer than 13 years of schooling	(2) More than 12 years of schooling	Relative demand shift (2) – (1)
1940s	2.02%	2.06%	0.04
1950s	1.13	2.00	0.87
1960s	1.16	2.18	1.02
1970s	1.61	1.82	0.21

Notes. A demand index in year t for schooling level j (D_{jt}) is constructed as a weighted average of the proportion of the white, male labor force who attained a particular level of schooling (j) by industry (i) in 1960. The weights (ρ_{it}), which change with time, are the employment shares in year t by industrial sector i . Thus, for each level of schooling (j) and time period (t), $D_{jt} = \sum_i \rho_{it} \cdot s_{ij}^{60}$. The (white male) labor force in year t of schooling level j , given the 1960 schooling weights, would be $L_t \cdot \bar{D}_{jt}$. The numbers given are the average annualized percentage change of this number.

Sources. U. S. Bureau of the Census [1943a, 1955, 1967, 1973, 1984].

workers with fewer than thirteen years of schooling for all four decades. But the difference between the two rates was never closer than it was in the 1940s. Indeed the relative demand shift—that is, the difference between columns (1) and (2) in Table IX—was 0.04 in the 1940s and 0.87 in the 1950s.

The supply of educated Americans is given in Table X, in which we consider only the 1940 to 1960 period. Across all age groups in the 1940s, the growth in high school graduates exceeded that for college graduates. But, using the two educational groups of Table IX, the relative supply of the college educated expanded greatly. Thus, wage compression in the 1940s between the lesser and more educated groups is, perhaps, no mystery.²¹ Relative demand stayed virtually constant, while relative supply increased.²² In the 1950s the supply of college-educated workers continued to increase faster than that of less-educated workers, while the growth in the relative demand for the more educated greatly exceeded that for the less educated. These demand effects continued into the 1960s, and the wage compression of the 1940s eventually began to unravel. It should be realized that the demand shifts we have computed are just a partial measure of total demand and in no decade were

21. Although we have highlighted the wage compression between high school graduates and college graduates, Table II also reports enough information so that a wage compression between the college educated and those without college is also evident for the 1940s.

22. See Reder [1962, pp. 264–65] for a similar conclusion about the role of increases in the relative supply of educated labor in causing wage compression in the 1940s.

TABLE X
DISTRIBUTION OF EDUCATIONAL ATTAINMENT, 1940-1960:
WHITE MALES, 25 TO 64 YEARS

	Years of schooling					
	< 8	8	9-11	12	13-15	16+
1940						
25-34	17.7%	23.8%	22.4%	21.2%	7.1%	7.8%
35-44	25.9	31.5	17.4	12.9	5.8	6.5
45-54	35.1	33.3	12.2	9.8	4.6	5.0
55-64	40.6	34.2	9.1	8.1	3.7	4.3
25-64	26.0	30.0	16.3	14.0	5.6	6.1
1950						
25-34	13.0	13.4	22.0	30.9	10.7	10.0
35-44	19.3	20.8	21.1	21.6	8.1	9.1
45-54	28.4	27.6	16.3	14.5	6.0	7.2
55-64	38.1	28.4	12.2	10.7	4.9	5.7
25-64	22.9	21.4	18.7	20.9	7.8	8.3
1960						
25-34	9.9	9.7	21.1	31.2	12.3	15.8
35-44	12.4	13.1	21.4	30.3	10.3	12.5
45-54	18.5	20.8	21.1	20.6	8.6	10.4
55-64	34.6	24.4	16.0	11.8	6.6	6.6
25-64	18.0	16.3	20.2	24.5	9.7	11.3

Source. Calculated from data in U. S. Bureau of the Census [1943b, 1953, 1964].

greater than the relative supply shift of educated workers. That is, according to the analysis, the relative price of skills would have decreased in every decade. But if the measured portion of the demand shift were a constant fraction of the total, the explanation we have just offered would be correct.

A major cause of the increased supply of college-educated workers between 1940 and 1950 was the GI Bill of Rights [Vatter, 1985, p. 136].²³ Many whose lives had been interrupted by military

23. The decreasing share of immigrants in the population, a consequence of restrictive legislation following World War I, also played a role because the pre-World War I waves of immigrants were less likely to have attended college than the native born. But it probably only accounts for, at most, 10 percent of the increase in the share of the college educated in the adult male population from 1940 to 1950. Further, immigrants in the 1930s and 1940s were disproportionately educated compared with the native population.

service took advantage of the bill's generous financial provisions for college attendance. The bill's impact is reflected in the number of college graduates per 100 persons aged 23, which greatly increased in 1948 [U. S. Bureau of the Census, 1975, p. 385].²⁴ This sudden and partly unanticipated rise in relative supply, as we have shown, helped depress the relative earnings of college-educated labor in the immediate aftermath of World War II. It should be emphasized that there need not be a single explanation for the compression in the wage structure during the 1940s. The extremely large reduction in the premium to college education, seen in Table II, could have been quite separate from reductions along other dimensions. That is, the impact of World War II and the NWLB preceded the increase in the supply of the college educated, and each could have affected different groups in the labor force, albeit reinforcing each other.

Not only did schooling levels change, but there are also good reasons to believe that the variance in school quality across the United States fell during the first half of the century. Urbanization, the centralization of school districts, the school bus, and the widespread creation of publicly financed high schools greatly expanded educational options. Steady growth of land-grant and other state-financed colleges and universities created new opportunities for high school graduates to continue their education. Regional variation in the quality of elementary and secondary education—as measured, for example, by the length of the school year or teacher/pupil ratios—was smaller in 1950 than in 1900 or 1930.²⁵ Compulsory education and child labor laws raised the opportunity costs (to parents) of keeping children out of school, and by the end of the 1920s virtually every state had passed

24. College attendance by veterans under the bill's provisions peaked in 1947, accounting for 49.2 percent of all persons (69.3 percent of males) enrolled in higher educational institutions in that year. In fact, enrollment by veterans can explain virtually all of the increase in total college enrollment between 1939 and 1947. Enrollment of veterans declined sharply after 1949 as a consequence of the conflict in Korea, causing a decline in total enrollment that lasted from 1950 to 1953 [Olson, 1974, p. 44].

25. The coefficients of variation, for 1900, 1930, and 1950, across states in the length of the school year and in pupil-teacher ratios (public schools only) were

	1900	1930	1950
Length of term	23.9	7.6	3.1
Pupil-teacher ratio	132.8	18.7	14.1

Source. U. S. Bureau of Education [1901], U. S. Department of the Interior [1932], U. S. Department of Health, Education and Welfare [1954].

legislation restricting child labor. The decrease in spatial variability in the quality of educational opportunities ought to have narrowed the wage spread within educational groups and may be responsible for some of the compression within groups that is observed.²⁶

III. CONCLUSION

During the past quarter century wages in the United States have become more unequally distributed. Yet for the previous quarter century wage compression, rather than wage stretching, was the rule. While some data indicate that the narrowing of the wage and income distributions began much earlier, evidence presented here suggests that returns to schooling may not have been reduced until the 1940s. The Great Compression, as we have called it, was primarily the result of a particular confluence of short-run events affecting the demand for labor and of institutional changes brought about by the war and the command economy that accompanied it. But the wage structure did not immediately rebound to its prewar level, as it had after a similar decrease in skill ratios during World War I. Nor did the wage structure by industry stop narrowing when World War II ended. Rather, compression was observed at the industry level during the period from the mid-1940s to the early-1950s, and compression occurred in both the upper and lower tails of the distribution. Several factors served to reinforce the immediate effects of the war. The relative demand for less-educated workers increased during the 1940s and 1950s, and a rising minimum wage continued to pull up the bottom of the wage distribution. The American labor movement was never stronger than in the 1950s; and unions, it has been claimed, were strongly in favor of a compressed wage structure, at least for a while. Finally, and perhaps of most importance, increases in the supply of educated labor served to depress the price of skilled labor and retained, for some time, the egalitarian impact of the Great Compression.

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26. Card and Krueger [1992] demonstrate that long-term increases in school quality (e.g., teacher/pupil ratios) have a positive effect on earnings by raising the rate of return to schooling.

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